NARSIMHA REDDY ENGINEERING COLLEGE
UGC AUTONOMOUS INSTITUTION
Maisammaguda (V), Kompally - 500100, Secunderabad, Telangana State, India
UGC - Autonomous Institute
Accredited by NBA & NAAC with 'A' Grade
Approved by AICTE
Permanently affiliated to JNTUH

PROJECT BASED LEARNING

ON

RECTANGULAR WATER TANK DESIGN BY USING AUTOCAD SOFTWARE

BY

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(26X05A0108)

UNDER THE GUIDANCE OF

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NARSIMHA REDDY ENGINEERING COLLEGE (UGC-AUTONOMOUS)

APPROVED BY AICTE, AFFILIATED TO JNTUH, ACCREDITED BY NBA & NAAC WITH A- GRADE)

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500100.(2026-2027)

ABSTRACT

The project entitled “Design of Rectangular Water Tank Using AutoCAD Software” focuses on the planning, designing, and drafting of a rectangular water storage tank using AutoCAD software. Water tanks play an important role in storing and supplying water for residential, commercial, and industrial purposes. The main objective of this project is to develop an accurate and efficient design of a rectangular water tank by considering its dimensions, storage capacity, structural components, and construction requirements.

In this project, AutoCAD software is used to prepare detailed 2D drawings of the rectangular water tank, including plan, elevation, and sectional views with proper dimensions and annotations. The design considers important components such as the base slab, side walls, top slab (if required), and water storage capacity. Basic structural design considerations, including water pressure, wall thickness, and stability, are also taken into account.

The use of AutoCAD helps to improve drawing accuracy, reduce manual errors, save time, and provide a clear visualization of the proposed structure. This project provides practical knowledge of computer-aided drafting and the basic principles of water tank design. Thus, the proposed design aims to provide a simple, economical, and efficient solution for water storage while enhancing technical skills in civil engineering design and drafting.

1. INTRODUCTION

Water is one of the most essential natural resources required for human life and various daily activities. Proper storage of water is important to ensure a continuous and sufficient water supply for residential, commercial, agricultural, and industrial purposes. A rectangular water tank is one of the commonly used structures for storing water due to its simple shape, easy construction, and efficient utilization of available space. The design of a rectangular water tank involves determining its length, breadth, depth, wall thickness, and storage capacity based on the required quantity of water. Proper design is necessary to ensure the safety, stability, durability, and efficient functioning of the tank.

In this project, AutoCAD software is used for designing and preparing the detailed drawings of a rectangular water tank. AutoCAD is a computer-aided design software that helps in preparing accurate engineering drawings with proper dimensions and measurements. Using this software, different views such as plan, elevation, and sectional views of the water tank can be developed easily. It also helps in reducing manual errors, saving time, and improving the clarity of engineering drawings. The main objective of this project is to understand the basic principles of rectangular water tank design and gain practical knowledge of computer-aided drafting. This project also provides an opportunity to improve technical skills and understand the importance of accurate planning and designing in the field of civil engineering. The proposed design aims to provide a simple, economical, and efficient solution for water storage.

2. DRAWING-1

Water tank plan

Tank Details

ITEM	DIMENSION
• Tank Type	RCC Overhead Water Tank
• Capacity	10,000 Litres
• Internal Length	3000 mm
• Internal Width	3000 mm
• Internal Depth	1500 mm
• Wall Thickness	200 mm
• Bottom Slab	200 mm
• Top Slab	200 mm

Drawing Includes

- Tank Plan
- Wall Thickness
- Inlet Pipe
- Outlet Pipe
- Overflow Pipe
- Washout pipe
- Manhole
- Vent pipe
- Dimensions

3. DRAWING-2

Front elevation

Elevation Details

Component	**Dimension**
Staging Height	6000mm
Column Size	300x300mm
Number of Columns	4
Brace Beam	230x230mm
Top Slab	150 mm

Elevation Includes

- RCC column
- Brace beams
- Water tank
- Ladder
- Manhole
- Ground level
- Overall height

4. DRAWING-3

Section A-A

Structural Details

Component	**Size**
Wall Thickness	200mm
Bottom Slab	200mm
Top Slab	150mm
Base Beam	300 x 450 mm
Column	300 x 300 mm

Reinforcement

- Wall steel = 12mm @150 mm c/c
- Slab Steel = 10mm @ 150mm c/c
- Column = 8 Nos 16mm O
- Strrups = 8 mm @ 150 mm c/c

Section Includes

- Water Level
- Bottom Slab
- Wall Reinforement
- Roof Slab
- Manhole
- Ladder
- Staging
- Foundation

5. DRAWING DETAILING WITH MEASUREMENTS

Tank

- Length=3000mm
- Width=3000mm
- Height=1500mm

Structural Members

- Wall = 200 mm
- Bottom Slab = 200 mm
- Roof Slab = 150 mm
- Column = 300 x 300 mm
- Brace Beam = 230 x 230 mm

Foundation

- Footing = 1800 x 1800 x 450 mm
- PCC = 100 mm

6. AUTOCAD DRAWING TOOLS USED

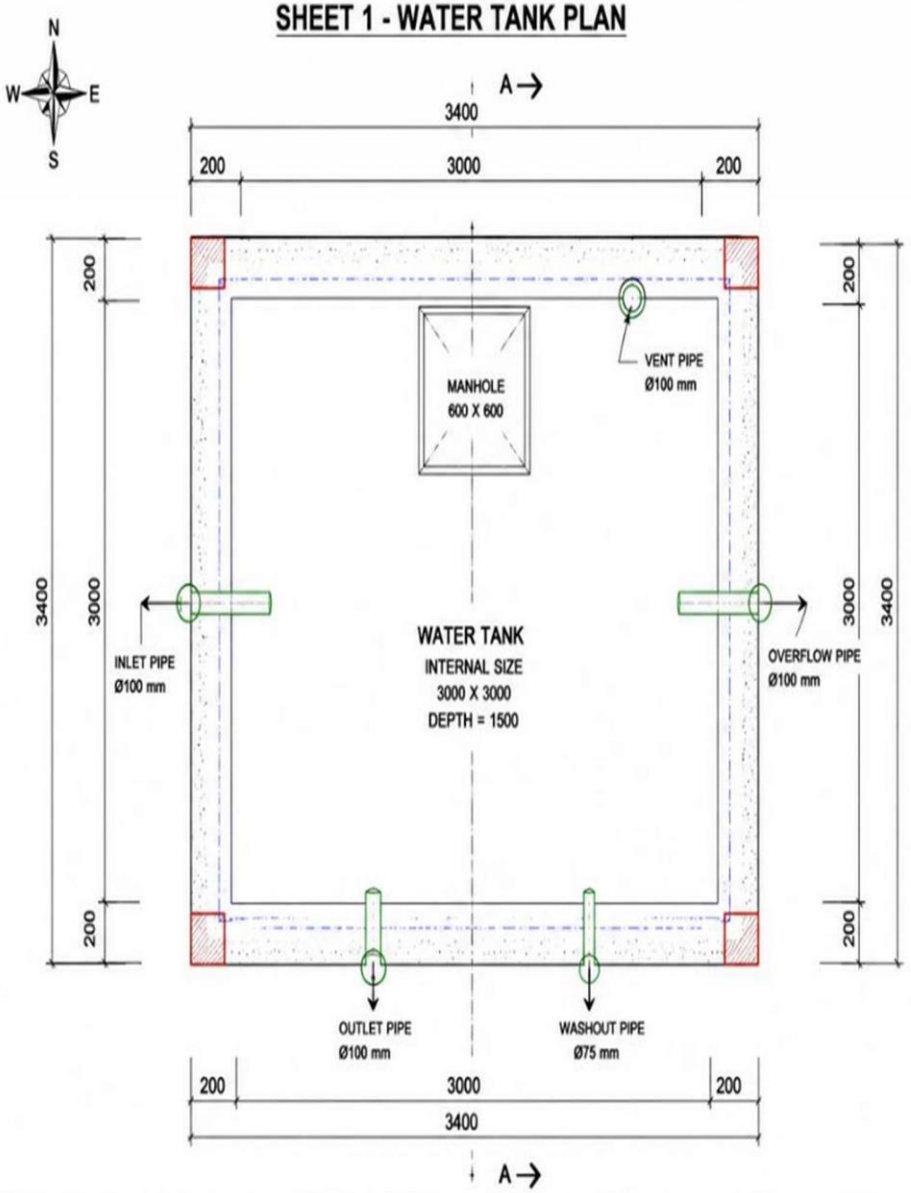
DRAWING TOOLS

- LINE
- POLYLINE
- RECTANGLE
- CIRCLE
- ARC
- POLYGON
- HATCH
- TEXT
- MTEXT
- DIMENSION
- LEADER
- TABLE

LAYER TOOLS

- LAYER PROPERTIES
- LINE TYPE
- LINE WEIGHT
- COLOUR MANAGEME

SHEET 1 - WATER TANK PLAN



WATER TANK DETAILS	
CAPACITY	10,000 LITRES
INTERNAL SIZE	3000 X 3000 mm
INTERNAL DEPTH	1500 mm
WALL THICKNESS	200 mm
BOTTOM SLAB THICKNESS	200 mm
TOP SLAB THICKNESS	150 mm
FREE BOARD	300 mm
CONCRETE GRADE	M25
STEEL GRADE	Fe500

OPENING SCHEDULE			
MARK	DESCRIPTION	SIZE (mm)	REMARKS
MH	MANHOLE	600 X 600	CLEAR OPENING
VP	VENT PIPE	Ø100	...
IP	INLET PIPE	Ø100	...
OP	OVERFLOW PIPE	Ø100	...
WP	WASHOUT PIPE	Ø75	...

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETERS.
2. DO NOT SCALE THE DRAWING.
3. PROVIDE 20 mm WATER PROOF PLASTER ON INTERNAL SURFACES.
4. PROVIDE ADEQUATE WATER PROOFING TREATMENT.
5. ALL RCC WORK SHALL BE IN ACCORDANCE WITH IS 456:2000.

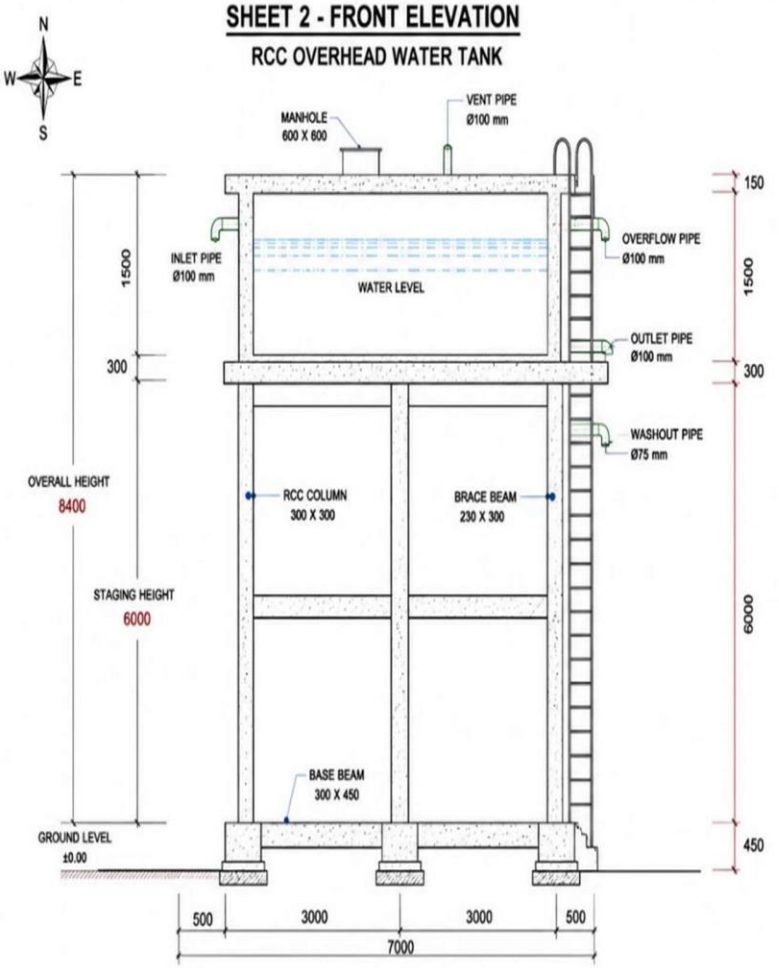
LEGEND

- COLUMN
- RCC WALL
- PIPE
- WATER LINE

PROJECT : OVERHEAD WATER TANK	SCALE : 1:50	NORTH	
DRAWING : WATER TANK PLAN	DATE : 22-05-2024		DRAWING NO. WT-01
DRAWN BY : CIVIL ENGINEER			

SHEET 2 - FRONT ELEVATION

RCC OVERHEAD WATER TANK



ELEVATION DETAILS	
CAPACITY	10,000 LITRES
INTERNAL SIZE	3000 X 3000 mm
INTERNAL DEPTH	1500 mm
FREE BOARD	300 mm
TOTAL TANK DEPTH	1800 mm
STAGING HEIGHT	6000 mm
OVERALL HEIGHT	8400 mm

MEMBER DETAILS		
MEMBER	SIZE	REINFORCEMENT
COLUMN	300 X 300	8 - 16mmØ 8mmØ @150c/c
BRACE BEAM	230 X 300	4 - 16mmØ 8mmØ @150c/c
BASE BEAM	300 X 450	4 - 16mmØ (TOP) 4 - 20mmØ (BOTTOM) 8mmØ @150c/c

MATERIAL DETAILS	
CONCRETE GRADE	M25
STEEL GRADE	Fe500

NOTES :-

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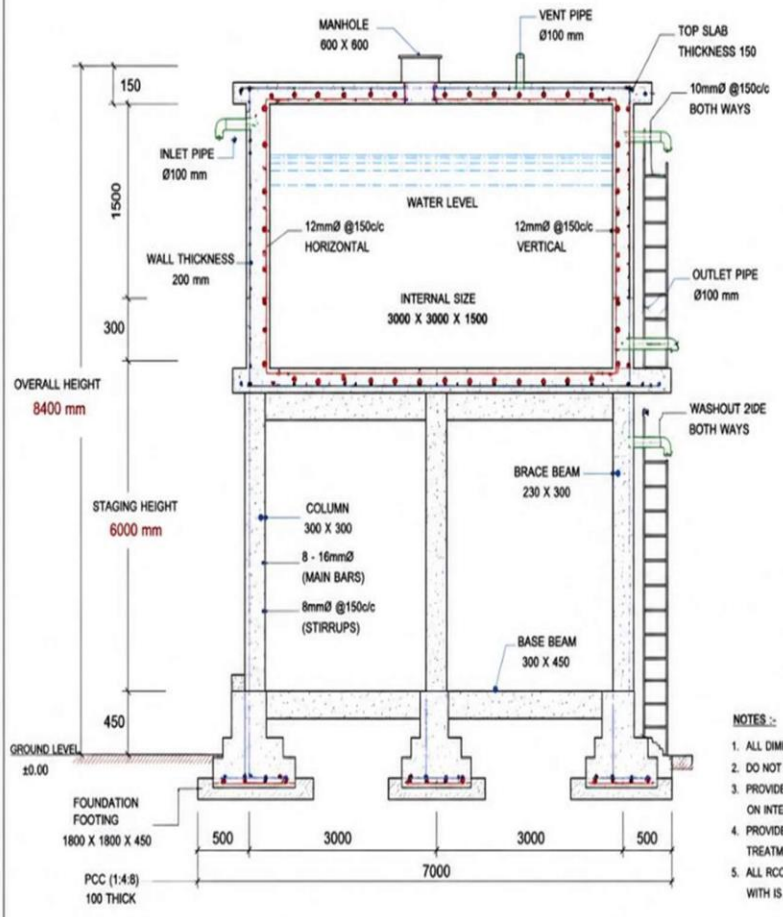
LEGEND

- PIPE
- RCC CONCRETE
- WATER LEVEL

PROJECT : OVERHEAD WATER TANK	SCALE : 1:50	NORTH	
DRAWING : FRONT ELEVATION	DATE : 22-05-2024		DRAWING NO. WT-02
DRAWN BY : CIVIL ENGINEER			

SHEET 3 - SECTION A-A

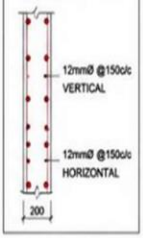
RCC OVERHEAD WATER TANK



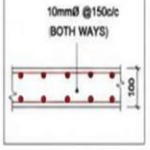
REINFORCEMENT DETAILS		
MEMBER	THICKNESS / SIZE	REINFORCEMENT
WALL	200 mm	12mmØ @150c/c VERTICAL & HORIZONTAL
BOTTOM SLAB	200 mm	10mmØ @150c/c BOTH WAYS
TOP SLAB	150 mm	10mmØ @150c/c BOTH WAYS

MATERIAL DETAILS	
CONCRETE GRADE	M25
STEEL GRADE	Fe500
WATER PROOFING	20 mm THICK WATER PROOF PLASTER
CLEAR COVER	WALLS & SLAB = 25 mm COLUMN = 40 mm
FREE BOARD	300 mm

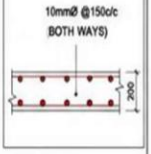
TANK WALL SECTION



ROOF SLAB SECTION



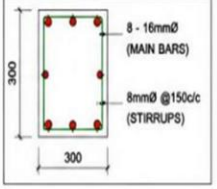
BOTTOM SLAB SECTION



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COLUMN SECTION (300 X 300)



PROJECT : OVERHEAD WATER TANK	SCALE : 1:50	NORTH	
DRAWING : SECTION A-A	DATE : 22-05-2024		DRAWING NO. WT-03
DRAWN BY : CIVIL ENGINEER			

7. MODIFY TOOLS USED

- MOVE
- COPY
- MIRROR
- OFFSET
- TRIM
- EXTEND
- ROTATE
- SCALE
- STRETCH
- ARRAY
- FILLET
- CHAMFER
- BREAK
- EXPLODE
- JOIN

8. TECHNOLOGY USED SOFTWARE

- AUTOCAD 2024 operating system
- Windows 10/windows 11 design standard
- IS 800:2007(general construction in steel)
- IS 875 (LOADING STANDARDS)
- IS 456:2000 (concrete)
- National building code (NBC) 2016 hardware
- Intel core i5 processor or above
- 8 GB RAM
- 512 GB SSD
- Dedicated graphics card (recommended)

9. APPLICATIONS

- Residential Buildings
- Apartments
- Schools
- Hospitals
- Industrial Plants
- Commeercial Complexes
- Municipal Water Supply Systems
- Rural Water Supply Projects

10. ADVANTAGES

- Safe Water Storage
- Durable RCC Construction
- Easy Maintanance
- Leak-resistance Design
- Accurate drafting using AutoCAD
- Proffesional Construction Drawings
- Better Stuctural safety
- Easy Future Modifications

11. CONCLUSION

The project “Design of Rectangular Water Tank Using AutoCAD Software” helped us understand the basic concepts and procedures involved in designing a water storage tank. Through this project, we gained practical knowledge of preparing 2D drawings, including plan, elevation, and sectional views using AutoCAD software. We also understood the importance of selecting proper dimensions, storage capacity, wall thickness, and structural components for a safe and efficient design. The use of AutoCAD made the drawing process easier, faster, and more accurate while reducing manual errors. This project improved our technical knowledge, drafting skills, and understanding of civil engineering design. Overall, the project provided valuable practical experience in using computer-aided design software for preparing engineering drawings and developing a simple, economical, and efficient rectangular water tank design.